CS340 Final Project: Bias Mitigation in ACSEmployment Prediction

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Abstract

This report evaluates gender fairness in employment prediction on the ACSEmployment dataset. We first benchmark a standard neural network with TFMA and five extended fairness metrics, then apply adversarial debiasing, an Equalized Odds loss, and sample re-weighting. The mitigated model improves all five extended fairness metrics, mainly by reducing predictive parity and AUC gaps. The improvement comes with a clear performance trade-off: recall increases, while precision, accuracy, and AUC decrease.

1 Dataset and Benchmark Analysis

1.1 Dataset Overview

We use the ACSEmployment dataset from the Folktables project, which contains American Community Survey (ACS) data for employment prediction. The dataset includes:

- **Geographic Coverage:** California (CA) and Texas (TX), 2018
- **Evaluation Set Size:** 129,383 individuals in the held-out split
- **Target Variable:** EMPLOYED (binary: employed vs. not employed)
- **Sensitive Attribute:** SEX (Female vs. Male)
- **Features:** Demographic and socioeconomic attributes (age, education, marital status, etc.)
- **Excluded Features:** RELP (relationship) removed to reduce proxy bias

Class Distribution:

- Positive class (employed): $\sim$45.4%
- Negative class (not employed): $\sim$54.6%

Gender Distribution:

- Female: $\sim$50.8%
- Male: $\sim$49.2%

1.2 Bias Evaluation and Metrics

We evaluate bias using two complementary sets of metrics:

1. TFMA (TensorFlow Model Analysis) Metrics (13 metrics):
Standard classification metrics evaluated overall and per gender group:

- **Performance:** AUC, binary accuracy, example count
- **Confusion Matrix Rates:** TPR, FPR, TNR, FNR, precision, recall
- **Error Rates:** False discovery rate, false omission rate
- **Prediction Rates:** Positive rate, negative rate

Table 1: Base Model Performance Metrics by Sex

Metric	Female	Male	Overall
Sample Size	65,819	63,564	129,383
AUC	0.8713	0.9244	0.9016
Binary Accuracy	0.7889	0.8527	0.8202
Positive Rate	0.4668	0.5253	0.4956
Negative Rate	0.5332	0.4747	0.5044
Precision	0.7216	0.8267	0.7763
Recall (TPR)	0.8059	0.8853	0.8481
True Negative Rate	0.7767	0.8213	0.7971
False Positive Rate	0.2233	0.1787	0.2029
False Negative Rate	0.1941	0.1147	0.1519
False Discovery Rate	0.2784	0.1733	0.2237
False Omission Rate	0.1521	0.1186	0.1366

2. Extended Fairness Metrics (5 metrics):
Specialized metrics for quantifying bias:

1. **Disparate Impact:** $\frac{\mathrm{PR}_{\mathrm{female}}}{\mathrm{PR}_{\mathrm{male}}}$ (fair if $0.8 \leq$ ratio ≤ 1.25)
2. **Average Odds:** $\frac{|\mathrm{TPR}_f - \mathrm{TPR}_m| + |\mathrm{FPR}_f - \mathrm{FPR}_m|}{2}$ (fair if < 0.1)
3. **Predictive Parity:** $|\mathrm{Precision}_f - \mathrm{Precision}_m|$ (fair if < 0.1)
4. **NPV Equality:** $|\mathrm{NPV}_f - \mathrm{NPV}_m|$ (fair if < 0.1)
5. **Performance Gaps (AUC):** $|\mathrm{AUC}_f - \mathrm{AUC}_m|$ (fair if < 0.05)

1.3 Observations on Baseline Bias

The baseline model (3-layer neural network with standard training) exhibits the following bias patterns:

The baseline is not uniformly unfair: it satisfies Disparate Impact, Average Odds, and NPV Equality. The remaining gaps are concentrated in prediction reliability and ranking performance. Female precision is 0.7216, compared with 0.8267 for male, producing a Predictive Parity gap of 0.1051, slightly above the 0.10 threshold. The AUC gap is also above threshold (0.0531), with lower AUC for the female group.

The error profile shows why mitigation is needed. The female group has both a higher FPR (0.2233 vs. 0.1787) and a higher FNR (0.1941 vs. 0.1147), meaning errors are less balanced across gender groups. Therefore, the mitigation objective should target error-rate parity and representation bias, not only positive prediction rates.

Table 2: Base Model Fairness Metrics Summary

Fairness Metric	Value/Difference	Threshold	Status
Disparate Impact (80% Rule)			
Female Positive Rate	0.4668		
Male Positive Rate	0.5253		
Ratio	0.8886	≥ 0.80	Fair
Average Odds Difference	0.0620	≤ 0.10	Fair
Predictive Parity			
Female Precision	0.7216		
Male Precision	0.8267		
Difference	0.1051	≤ 0.10	Unfair
NPV Equality (Cond. Use Accuracy)			
Female NPV	0.8479		
Male NPV	0.8814		
Difference	0.0335	≤ 0.10	Fair
Performance Gaps (AUC)			
Female AUC	0.8713		
Male AUC	0.9244		
AUC Gap	0.0531	≤ 0.05	Unfair

2 Bias Mitigation Methodology

2.1 Design Idea and Rationale

Our bias mitigation strategy combines three complementary techniques to address fairness concerns while maintaining predictive performance:

1. Adversarial Debiasing: We employ a gradient reversal layer (GRL) to prevent the model from learning gender-specific patterns. The main classifier predicts employment status, while an adversarial classifier attempts to predict gender from the learned representations. The GRL reverses gradients during backpropagation, forcing the main classifier to learn gender-invariant features.

2. Equalized Odds Constraint: We explicitly minimize the difference in False Negative Rate (FNR) and False Positive Rate (FPR) between gender groups by adding a fairness loss term:

$$\mathcal{L}_{fair} = |\text{FNR}_{\text{female}} - \text{FNR}_{\text{male}}| + |\text{FPR}_{\text{female}} - \text{FPR}_{\text{male}}| \tag{1}$$

3. Sample Re-weighting: We apply gender-balanced weights to ensure equal representation during training, with additional boosting ($1.5\times$) for positive examples to address class imbalance.

Rationale: This multi-faceted approach addresses bias at different levels:

- Adversarial debiasing discourages gender information in learned representations
- Equalized Odds directly targets FNR and FPR gaps across gender groups
- Sample re-weighting reduces group and class imbalance during optimization

2.2 Implementation Details

Model Architecture:

- **Input Layer:** Feature normalization using batch statistics

- **Main Classifier:** Two hidden layers (64 units each) with Batch Normalization, ReLU activation, and Dropout (0.2), followed by a 32-unit layer and sigmoid output
- **Adversarial Branch:** Gradient Reversal Layer ($\lambda_{adv} = 0.5$) followed by a 16-unit hidden layer and sigmoid output for gender prediction

Loss Function:

$$\mathcal{L}_{total} = \mathcal{L}_{main} + \lambda_{adv} \cdot \mathcal{L}_{adv} \tag{2}$$

where:

$$\mathcal{L}_{main} = \mathcal{L}_{BCE}(y, \hat{y}) + \lambda_{fair} \cdot \mathcal{L}_{fair} \tag{3}$$

with $\lambda_{adv} = 0.5$ and $\lambda_{fair} = 0.5$.

Training Configuration:

- Optimizer: Adam with learning rate 10^{-3}
- Batch size: 256
- Epochs: 25
- Gradient clipping: Global norm clipping at 1.0
- Early stopping: Based on validation AUC

Sample Weights:

$$w_i = \frac{n}{2 \cdot n_{\text{gender}}} \times \begin{cases} 1.5 & \text{if } y_i = 1 \\ 1.0 & \text{if } y_i = 0 \end{cases} \tag{4}$$

2.3 Evaluation of the Mitigated Model

We evaluate the mitigated model using the same 13 TFMA metrics and 5 extended fairness metrics as the baseline.

Table 3: Fair Model Performance Metrics by Sex

Metric	Female	Male	Overall
Sample Size	65,819	63,564	129,383
AUC	0.8615	0.8994	0.8784
Binary Accuracy	0.7665	0.8188	0.7922
Positive Rate	0.5835	0.6064	0.5947
Negative Rate	0.4165	0.3936	0.4053
Precision	0.6580	0.7551	0.7066
Recall (TPR)	0.9186	0.9334	0.9265
True Negative Rate	0.6572	0.7085	0.6807
False Positive Rate	0.3428	0.2915	0.3193
False Negative Rate	0.0814	0.0666	0.0735
False Discovery Rate	0.3420	0.2449	0.2934
False Omission Rate	0.0816	0.0830	0.0823

Results Summary:

Compared with the baseline, the fair model raises overall recall from 0.8481 to 0.9265 and reduces the false negative rate from 0.1519 to 0.0735. This is important in an employment setting because false negatives correspond to qualified individuals being predicted as not employed.

The fairness gains are visible across the extended metrics. Predictive Parity decreases (0.1051 $\rightarrow$ 0.0971), NPV Equality becomes nearly equal (0.0335 $\rightarrow$ 0.0014), the AUC gap decreases

Table 4: Fair Model Fairness Metrics Summary

Fairness Metric	Value/Difference	Threshold	Status
Disparate Impact (80% Rule)			
Female Positive Rate	0.5835		
Male Positive Rate	0.6064		
Ratio	0.9622	≥ 0.80	Fair
Average Odds Difference	0.0330	≤ 0.10	Fair
Predictive Parity			
Female Precision	0.6580		
Male Precision	0.7551		
Difference	0.0971	≤ 0.10	Fair
NPV Equality (Cond. Use Accuracy)			
Female NPV	0.9184		
Male NPV	0.9170		
Difference	0.0014	≤ 0.10	Fair
Performance Gaps (AUC)			
Female AUC	0.8615		
Male AUC	0.8994		
AUC Gap	0.0378	≤ 0.05	Fair

Table 5: Extended Fairness Metric Comparison

Metric	Base Model	Fair Model	Change
Disparate Impact Ratio	0.8886	0.9622	Closer to 1
Average Odds Difference	0.0620	0.0330	-0.0290
Predictive Parity Gap	0.1051	0.0971	-0.0080
NPV Equality Gap	0.0335	0.0014	-0.0321
AUC Gap	0.0531	0.0378	-0.0153

(0.0531 $\rightarrow$ 0.0378), and Average Odds decreases (0.0620 $\rightarrow$ 0.0330). Disparate Impact also moves closer to parity, from 0.8886 to 0.9622. Across the complete evaluation set, 9 metrics improve in total: all 5 extended fairness metrics and 4 TFMA metrics (recall, true positive rate, false negative rate, and false omission rate).

3 Comparative Study and Research

3.1 Performance Comparison

The mitigated model is not uniformly better on predictive metrics. Overall AUC decreases from 0.9016 to 0.8784, and binary accuracy decreases from 0.8202 to 0.7922. At the 0.5 threshold, however, recall improves from 0.8481 to 0.9265, while precision decreases from 0.7763 to 0.7066.

This pattern indicates a threshold-level shift toward more positive predictions: the positive rate increases from 0.4956 to 0.5947. As a result, the model rejects fewer positive cases but also produces more false positives, reflected in the FPR increase from 0.2029 to 0.3193.

3.2 Fairness-Accuracy Trade-off Analysis

The trade-off is favorable only under a clear cost preference. In employment prediction, a false negative can represent a qualified candidate being incorrectly rejected, so the recall and FNR improvements are meaningful. At the same time, the model sacrifices AUC, accuracy, precision, and true negative rate. The correct interpretation is therefore targeted fairness improvement with measurable performance costs, not universal performance improvement.

3.3 Improvement Strategies

To preserve predictive utility while reducing fairness gaps, we employed several strategies:

1. Architectural Improvements:

- **Batch Normalization:** Stabilizes training and improves generalization
- **Dropout (0.2):** Prevents overfitting to majority group patterns
- **Gradient Clipping:** Ensures stable training with multiple loss terms

2. Training Strategies:

- **Early Stopping:** Prevents overfitting by monitoring validation AUC
- **Sample Re-weighting:** Balances gender representation and addresses class imbalance
- **Multi-objective Optimization:** Jointly optimizes for accuracy and fairness

3. Loss Function Design:

- **Weighted BCE:** Incorporates sample weights for balanced learning
- **Equalized Odds Term:** Explicitly minimizes FNR and FPR gaps
- **Adversarial Loss:** Removes gender information from representations

4 Conclusion

This project shows that fairness constraints can substantially improve group-level equity in employment prediction, but the improvement is not free. The fair model improves all five extended fairness metrics and retains usable predictive performance (AUC: 0.8784), while accepting lower accuracy and precision.

Key Findings:

1. **Fairness is achievable:** Through adversarial debiasing, Equalized Odds constraints, and sample re-weighting, the model improves all five extended fairness metrics.
2. **Performance cost is measurable:** Accuracy decreases from 0.8202 to 0.7922 and AUC from 0.9016 to 0.8784, while recall increases from 0.8481 to 0.9265.
3. **Predictive Parity matters:** Predictive Parity moves from unfair to fair (0.1051 $\rightarrow$ 0.0971), improving reliability balance across gender groups.
4. **Multi-level mitigation is useful:** Representation learning, loss design, and sample weighting address different sources of bias.

Practical Implications:

For real-world deployment of employment prediction systems:

- Fairness should be a design requirement, not an afterthought
- Higher recall and lower FNR mean fewer positive cases are incorrectly rejected
- Lower precision and higher FPR require monitoring before deployment
- Regular fairness audits across multiple metrics are essential for maintaining equitable systems